

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103000900 - Ethics And Law For Data Science

DEGREE PROGRAMME

10BA - Master Universitario En Ciencia De Datos

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

Index

Learning guide

1. Description.....	1
2. Faculty.....	1
3. Skills and learning outcomes	2
4. Brief description of the subject and syllabus.....	2
5. Schedule.....	4
6. Activities and assessment criteria.....	7
7. Teaching resources.....	9
8. Other information.....	10

1. Description

1.1. Subject details

Name of the subject	103000900 - Ethics And Law For Data Science
No of credits	3 ECTS
Type	Compulsory
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	10BA - Master Universitario en Ciencia de Datos
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Victor Rodriguez Doncel (Subject coordinator)	D3205	victor.rodriguez@upm.es	M - 10:00 - 13:00 Tu - 10:00 - 13:00
Asuncion De Maria Gomez Perez		asunciondemaria.gomez@upm.es	Sin horario.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CB08 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios

CECD09 - Capacidad para actuar con los principios éticos y legales relacionados con la manipulación de datos según el ámbito de aplicación

CG10 - Apreciación de los límites del conocimiento actual y de la aplicación práctica de la última tecnología

3.2. Learning outcomes

RA38 - Ability to assess the societal, legal and ethical impact of Artificial Intelligence and data processing projects

RA37 - Knowledge of the European and national legal framework of AI and data processing

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

Data scientists and AI professionals will face legal and ethical issues during their careers. Designing, deploying, or using AI systems and data-intensive applications requires knowing and abiding by the law (legal compliance) and being aware of the technology's impact and acting accordingly (ethical responsibility).

In this course, students will learn about European Union policies and regulations related to AI and data. Specifically, they will cover the basics of the EU AI Regulation and the AI Liability Directive, the General Data Protection Regulation (GDPR), the Data Governance Act, the Data Act, and some aspects of copyright law, including the Database Directive. The course will also cover open data policies in Europe.

4.2. Syllabus

1. Introduction. Overview of issues raised by Artificial Intelligence.
2. Introduction to ethics.
3. Professional ethics and deontological codes.
4. Methodologies for the ethical assessment of AI and data projects.
5. Bias and explainability of AI systems
6. Information ethics. The future of AI.
7. Privacy and data protection. GDPR.
8. Regulation of AI.
9. Regulation of data.
10. Intellectual Property Rights
11. Copyright and data licenses.
12. Identity and Security.

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Lecture. Duration: 02:00 Lecture Lecture Duration: 01:00 Problem-solving class			
2	Lecture. Duration: 01:30 Design thinking Lecture Duration: 01:00 Problem-solving class Questionnaire Duration: 00:30 Additional activities			
3	Lecture Duration: 02:00 Lecture Lecture Duration: 01:00 Problem-solving class			
4	Lecture Duration: 01:30 Lecture Lecture Duration: 01:00 Problem-solving class Questionnaire Duration: 00:30 Additional activities			Questionnaire (10%) Written test Progressive assessment Presential Duration: 00:30
5	Lecture Duration: 02:00 Lecture Lecture Duration: 01:00 Problem-solving class			

6	Lecture Duration: 01:30 Lecture Lecture Duration: 01:00 Problem-solving class Questionnaire Duration: 00:30 Additional activities			Questionnaire (10%) Written test Progressive assessment Presential Duration: 00:30
7	Lecture Duration: 01:30 Lecture Lecture Duration: 01:00 Problem-solving class Questionnaire Duration: 00:30 Additional activities			Intermediate work group delivery (5%) Group work Progressive assessment Not Presential Duration: 00:00 Questionnaire (5%) Written test Progressive assessment Presential Duration: 00:31
8	Work presentation Duration: 02:00 Additional activities Lecture Duration: 01:00 Problem-solving class			Work presentation (10%) Individual presentation Progressive assessment Presential Duration: 02:00 Final group work delivery (15%) Group work Progressive assessment Presential Duration: 00:00
9				Global exam Written test Progressive assessment Presential Duration: 02:00
10				
11				
12				
13				
14				
15				
16				
17				Exam Written test Global examination Presential Duration: 02:00 Work delivery including video presentation Individual work Global examination Not Presential Duration: 00:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27

hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
4	Questionnaire (10%)	Written test	Face-to-face	00:30	10%	0 / 10	CG10 CB08 CECD09
6	Questionnaire (10%)	Written test	Face-to-face	00:30	10%	0 / 10	CG10 CB08 CECD09
7	Intermediate work group delivery (5%)	Group work	No Presential	00:00	5%	0 / 10	CECD09 CG10 CB08
7	Questionnaire (5%)	Written test	Face-to-face	00:31	10%	0 / 10	CG10 CB08 CECD09
8	Work presentation (10%)	Individual presentation	Face-to-face	02:00	10%	4 / 10	CG10 CB08 CECD09
8	Final group work delivery (15%)	Group work	Face-to-face	00:00	15%	4 / 10	CECD09 CG10 CB08
9	Global exam	Written test	Face-to-face	02:00	40%	4 / 10	CG10 CB08 CECD09

6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Exam	Written test	Face-to-face	02:00	60%	4 / 10	CG10 CB08 CECD09
17	Work delivery including video presentation	Individual work	No Presential	00:00	40%	4 / 10	CG10 CB08 CECD09

6.1.3. Referred (re-sit) examination

No se ha definido la evaluación extraordinaria.

6.2. Assessment criteria

EVALUATION SYSTEM

There are three different evaluation activities:

- **QUESTIONNAIRES.** During the course, Moodle questionnaires will be open as in-class activities or homework tasks. (30%)
- **GROUP WORK.** This group work (4 students per group) will have an intermediate delivery (towards the middle of the course) and a final delivery. At the end of the course, there will be an oral presentation (possibly video). (30%)
- **GLOBAL EXAM.** This final exam will consist of two parts. First, a written test with no materials. Second, a Moodle questionnaire, where any online or offline material is allowed. All the competencies of the course will be evaluated in this exam. For this second part, a printed version will be also available for students with no electronic device. Each of the two parts needs to obtain at least a 4/10 in order to pass the course. (40%)

The group work and the global exam need to obtain at least 4/10. Each of the two parts in the global exam needs to be evaluated with at least 4/10 in order to pass the course.

EVALUATION ONLY THROUGH GLOBAL EXAM

Notwithstanding the global exam (60%), students will have to deliver the work, even if individually, and will have to record a video to replace the oral presentation (40%). Both parts require at least 4/10.

EXTRAORDINARY EXAM

Notwithstanding the global exam (60%), students will have to deliver the work, even if individually, and will have to record a video to replace the oral presentation (40%). Both parts require at least 4/10. The group work is the same as the one described before, no additional delivery is necessary if passed during the progressive evaluation.

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
Textbook	Bibliography	Quinn, M. J. (2017). Ethics for the information age. Pearson.
Moodle	Bibliography	A collection of readings will be made available through the moodle platform. Additional recommended bibliography will be also referenced from Moodle.

8. Other information

8.1. Other information about the subject

Language

The course is delivered in English.

You are advice to bring a laptop or tablet to the lessons